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تمرین ۲ سیستم عامل

بخش الف)

ابتدا با دستور nano فایل c را ایجاد میکنیم
به نام Ex2_1.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <fcntl.h>

int main() {
    int fd;
    char buffer[100];

    fd = open("sample.txt", O_CREAT | O_WRONLY, 0644);
    if (fd < 0) {
        perror("open");
        return 1;
    }

    write(fd, "Hello, Operating System!\n", 25);
    close(fd);

    printf("Process ID: %d\n", getpid());
    printf("Parent Process ID: %d\n", getppid());

    return 0;
}
```

سپس با دستور gcc -o program Ex2_1.c آن را کامپایل میکنیم

با دستور ./program آن را اجرا میکنیم

در نهایت دستورات:

strace ./program

strace -c ./program

System Call های اصلی شناسایی شده:

- 1 - openat() برای باز کردن فایل sample.txt با پارامترهای ایجاد و نوشتن
- 2 - write() برای نوشتن متن "Hello, Operating System!" در فایل
- 3 - close() برای بستن فایل پس از عملیات نوشتن
- 4 - getpid() دریافت شناسه پردازش جاری
- 5 - getppid() دریافت شناسه پردازش والد

```
Nov 15 21:01
parsadarsa@parsadarsa: ~/Desktop
parsadarsa@parsadarsa:~/Desktop$ nano Ex2_1.c
parsadarsa@parsadarsa:~/Desktop$ gcc -o program Ex2_1.c
parsadarsa@parsadarsa:~/Desktop$ ./program
Process ID: 9100
Parent Process ID: 3186
parsadarsa@parsadarsa:~/Desktop$ ./Ex2_1
bash: ./Ex2_1: No such file or directory
parsadarsa@parsadarsa:~/Desktop$ gcc -o program Ex2_1.c
parsadarsa@parsadarsa:~/Desktop$ ./program
Process ID: 9234
Parent Process ID: 3186
parsadarsa@parsadarsa:~/Desktop$ strace ./program
execve("./program", ["/.program"], 0xffff5a8d1b0 /* 40 vars */) = 0
brk(NULL) = 0x2e3fd85000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x2d3e0498000
fcntl(AT_FDCWD, "/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=64723, ...}) = 0
mmap(NULL, 64723, PROT_READ, MAP_PRIVATE, 3, 0) = 0x2d3e0488000
close(3) = 0
openat(AT_FDCWD, "/lib/aarch64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0\267\0\1\0\0\360\206\2\0\0\0\0", 832) = 832
fstat(3, {st_mode=S_IFREG|0755, st_size=1722920, ...}) = 0
mmap(NULL, 1892240, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_DENYWRITE, -1, 0) = 0x2d3e0291000
mmap(0x2d3e02a000, 1826704, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0x2d3e02a000
munmap(0x2d3e0291000, 61440) = 0
munmap(0x2d3e045e000, 39840) = 0
mprotect(0x2d3e0439000, 81920, PROT_NONE) = 0
mmap(0x2d3e0440000, 20480, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x19d000) = 0x2d3e0440000
mmap(0x2d3e0452000, 49640, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x2d3e0452000
close(3) = 0
set_tid_address(0x2d3e0498fb0) = 9246
set_robust_list(0x2d3e0498fb0, 24) = 0
rseq(0x2d3e0499600, 0x20, 0, 0xd428bc00) = 0
mprotect(0x2d3e044d000, 12288, PROT_READ) = 0
mprotect(0x2d3e07cffe00, 4096, PROT_READ) = 0
mprotect(0x2d3e049d000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
munmap(0x2d3e0488000, 64723) = 0
openat(AT_FDCWD, "sample.txt", O_WRONLY|O_CREAT, 0644) = 3
write(3, "Hello, Operating System!\n", 25) = 25
close(3) = 0
getpid() = 9246
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x80, 0), ...}) = 0
getrandom("\x62\x6c\x7a\x11\xdf\x16\xac\x5d", 8, GRND_NONBLOCK) = 8
brk(NULL) = 0x2e3fd85000
brk(0x2e3fd86000) = 0x2e3fd86000
write(1, "Process ID: 9246\n", 17) = 17
getppid() = 9243
write(1, "Parent Process ID: 9243\n", 24) = 24
exit_group(0) = ?
+++ exited with 0 +++
```

```
Nov 16 08:38
parsadarsa@parsadarsa: ~/Desktop
parsadarsa@parsadarsa:~/Desktop$ strace -c ./program1
strace: Cannot stat './program1': No such file or directory
parsadarsa@parsadarsa:~/Desktop$ strace -c ./program
=== File Operations ===
File content: This is a test file.
=== Directory Operation ===
Directory created successfully.
=== Process Operations ===
Child Process - PID: 4279, Parent PID: 4277
=== System Info ===
Current PID: 4279
Parent PID: 4277
User ID: 1000
Parent Process - PID: 4277, Child PID: 4279
=== System Info ===
Current PID: 4277
Parent PID: 4274
User ID: 1000
% time    seconds  usecs/call   calls   errors syscall
-----
24.55    0.000122      11        11         0   write
10.06    0.000050      12         4         0   mprotect
9.66     0.000048      12         4         0   openat
9.66     0.000048       8         6         0   mmap
8.65     0.000043      43         1         0   clone
6.24     0.000031      10         3         0   munmap
6.04     0.000030      30         1         0   mkdirat
5.43     0.000027       6         4         0   close
3.02     0.000015       5         3         0   brk
2.62     0.000013       6         2         0   read
2.01     0.000010      10         1         0   wait4
1.61     0.000008       4         2         0   getpid
1.41     0.000007       2         3         0   fstat
1.41     0.000007       7         1         0   set_robust_list
1.41     0.000007       7         1         0   prlimit64
1.41     0.000007       7         1         0   getrandom
1.41     0.000007       7         1         0   rseq
1.21     0.000006       6         1         0   getppid
1.21     0.000006       6         1         0   getuid
1.01     0.000005       5         1         0   set_tid_address
0.00     0.000000       0         1         1   faccessat
0.00     0.000000       0         1         0   execve
-----
100.00    0.000497       9        54         1 total
parsadarsa@parsadarsa:~/Desktop$
```

بخش ب)

مشابه مراحل قبل:

System Call های جدید اضافه شده:

- 1 - read() برای خواندن محتوا از فایل
- 2 - fork() برای ایجاد پردازش فرزند
- 3 - getuid() برای دریافت شناسه کاربری
- 4 - mkdir() برای ایجاد دایرکتوری جدید

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>
#include <string.h>

int main() {
    int fd;
    char buffer[100];

    fd = open("test.txt", O_CREAT | O_WRONLY, 0644);
    write(fd, "Hello from new program!\n", 24);
    close(fd);

    fd = open("test.txt", O_RDONLY);
    int bytes_read = read(fd, buffer, sizeof(buffer)-1);
    buffer[bytes_read] = '\0';
    close(fd);

    printf("File content: %s", buffer);

    printf("My PID: %d\n", getpid());
    printf("Parent PID: %d\n", getppid());

    return 0;
}
```

```

parsadarsa@parsadarsa:~/Desktop$ nano Ex2_2.c
parsadarsa@parsadarsa:~/Desktop$ gcc -o program2 Ex2_2.c
parsadarsa@parsadarsa:~/Desktop$ ./program2
File content: Hello from new program!
My PID: 4764
Parent PID: 4261
parsadarsa@parsadarsa:~/Desktop$ strace -o trace2_output.txt ./program2
File content: Hello from new program!
My PID: 4780
Parent PID: 4776
parsadarsa@parsadarsa:~/Desktop$ ^[[200~strace ./program
strace: command not found
parsadarsa@parsadarsa:~/Desktop$ strace ./program
execve("./program", ["../program"], 0xfffff564b960 /* 48 vars */) = 0
brk(NULL) = 0xca1eb1f73000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xf418a19f9000
faccessat(AT_FDCWD, "/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=64723, ...}) = 0
mmap(NULL, 64723, PROT_READ, MAP_PRIVATE, 3, 0) = 0xf418a19e9000
close(3) = 0
openat(AT_FDCWD, "/lib/aarch64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0\267\0\1\0\0\0\360\2\0\0\0\0"... , 832) = 832
fstat(3, {st_mode=S_IFREG|0755, st_size=172920, ...}) = 0
mmap(NULL, 1692240, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_DENYWRITE, -1, 0) = 0xf418a17f2000
mmap(0xf418a1000000, 1826704, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0xf418a1000000
munmap(0xf418a17f2000, 57344) = 0
munmap(0xf418a19e000, 8000) = 0
mprotect(0xf418a1999000, 81920, PROT_NONE) = 0
mmap(0xf418a19ad000, 20480, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x19d000) = 0xf418a19ad000
mmap(0xf418a19b2000, 49040, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0xf418a19b2000
close(3) = 0
set_tid_address(0xf418a19f9fb0) = 4804
set_robust_list(0xf418a19f9fc0, 24) = 0
rseq(0xf418a19fa000, 0x20, 0, 0xd420bc00) = 0
mprotect(0xf418a19ad000, 12288, PROT_READ) = 0
mprotect(0xca1ebcf9f000, 4096, PROT_READ) = 0
mprotect(0xf418a19fe000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
munmap(0xf418a19e9000, 64723) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
getrandom("\x40\x3e\x74\x2e\x93\xb6\x30\x9e", 8, GRND_NONBLOCK) = 8
brk(NULL) = 0xca1eb1f73000
brk(0xca1eb1f94000) = 0xca1eb1f94000
write(1, "=== File Operations ===\n", 24) === File Operations ===
) = 24
openat(AT_FDCWD, "test1.txt", O_WRONLY|O_CREAT, 0644) = 3
write(3, "This is a test file.\n", 21) = 21
close(3) = 0
openat(AT_FDCWD, "test1.txt", O_RDONLY) = 3
read(3, "This is a test file.\n", 99) = 21
write(1, "File content: This is a test fil...", 35)File content: This is a test file.
) = 35
close(3) = 0
write(1, "=== Directory Operation ===\n", 28) === Directory Operation ===
) = 28
mkdirat(AT_FDCWD, "test_dir", 0755) = 0
write(1, "Directory created successfully.\n", 32)Directory created successfully.
) = 32
write(1, "=== Process Operations ===\n", 27) === Process Operations ===
) = 27
clone(child_stack=NULL, flags=CLONE_CHILD_CLEARTID|CLONE_CHILD_SETTID|SIGCHLD, child_tidptr=0xf418a19f9fb0) = 4806
getpid() = 4804
Child Process - PID: 4806, Parent PID: 4804
=== System Info ===

```

```

Current PID: 4806
Parent PID: 4804
User ID: 1000
write(1, "Parent Process - PID: 4804, Chil...", 44)Parent Process - PID: 4804, Child PID: 4806
) = 44
wait4(-1, NULL, 0, NULL) = 4806
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=4806, si_uid=1000, si_status=0, si_utime=0, si_stime=0} ---
write(1, "=== System Info ===\n", 20) === System Info ===
) = 20
getpid() = 4804
write(1, "Current PID: 4804\n", 18)Current PID: 4804
) = 18
getppid() = 4801
write(1, "Parent PID: 4801\n", 17)Parent PID: 4801
) = 17
getuid() = 1000
write(1, "User ID: 1000\n", 14)User ID: 1000
) = 14
exit_group(0) = ?
+++ exited with 0 +++

```

```
parsadarsa@parsadarsa:~/Desktop$ strace -c ./program2
```

```
File content: Hello from new program!
```

```
My PID: 4821
```

```
Parent PID: 4818
```

% time	seconds	usecs/call	calls	errors	syscall
32.66	0.000387	387	1		execve
13.25	0.000157	39	4		write
11.73	0.000139	23	6		mmap
7.76	0.000092	23	4		mprotect
7.00	0.000083	20	4		openat
6.08	0.000072	24	3		munmap
5.23	0.000062	20	3		brk
3.29	0.000039	13	3		fstat
2.95	0.000035	8	4		close
2.19	0.000026	13	2		read
1.35	0.000016	16	1	1	faccessat
1.35	0.000016	16	1		prlimit64
1.35	0.000016	16	1		getrandom
1.27	0.000015	15	1		rseq
1.10	0.000013	13	1		set_tid_address
1.01	0.000012	12	1		set_robust_list
0.25	0.000003	3	1		getpid
0.17	0.000002	2	1		getppid
100.00	0.001185	28	42	1	total